LAB 19.3

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Batch-46

Task 1: Python → C++

Prompt

Convert a Python Student class into equivalent C++ code with constructor and methods.

Python Code

```
class Student:
  def __init__(self, name, age):
    self.name = name
    self.age = age

  def display(self):
    print(self.name, self.age)

s = Student("varun", 20)
s.display()
```

Output

varun 20

C++ Code

```
#include <iostream>
using namespace std;

class   Student
  {          string
  name;      int
  age;

public:
  Student(string n, int a)
    { name = n;
    age = a;
  }
```

```
    void display() {
        cout << name << " " << age;
    }
};

int main() {
    Student s("varun", 20);
    s.display();
}
```

Output

varun 20

Task 2: Java → Python (Prime)

Prompt

Convert a Java isPrime() method into a Python function.

Code

```
def isPrime(n):
    if n <= 1:
        return False
    for i in range(2, n):
        if n % i == 0:
            return False
    return True

print(isPrime(7))
print(isPrime(10))
```

Output

True
False

Task 3: Pseudocode → Python

Prompt

Convert sorting pseudocode into Python (bubble sort).

Code

```
arr = [5, 2, 9, 1]

for i in range(len(arr)):
    for j in range(len(arr)-i-1):
        if arr[j] > arr[j+1]:
            arr[j], arr[j+1] = arr[j+1], arr[j]

print(arr)
```

Output

```
[1, 2, 5, 9]
```

Task 4: SQL → Pandas

Prompt

Convert SQL query into Pandas: select name and salary where salary > 50000.

Code

```
import pandas as pd

data = {
    'name': ['A', 'B', 'C'],
    'salary': [40000, 60000, 70000]
}

df = pd.DataFrame(data)

result = df[df['salary'] > 50000][['name', 'salary']]
print(result)
```

Output

```
  name  salary
1    B   60000
2    C   70000
```

Task 5: Python → C (Search)

Prompt

Convert a Python linear search algorithm into C.

Python Code

```
arr = [1, 3, 5, 7]
key = 5

for i in range(len(arr)):
    if arr[i] == key:
        print("Found at index", i)
```

Output

Found at index 2

C Code

```
#include <stdio.h>

int main() {
    int arr[] = {1,3,5,7};
    int key = 5;

    for(int
        i=0;i<4;i++){ if(arr[
        i]==key){
            printf("Found at index %d", i);
        }
    }
}
```

Output

Found at index 2